

HALO: A Heterogeneity-Aware, Language-Conditioned IMU Foundation Model for Label-Efficient Human Activity Recognition

ABSTRACT

Inertial measurement units (IMUs) are ubiquitous in phones and watches, yet human activity recognition (HAR) still has no foundation model that developers can readily build on: one pretrained model that ingests whatever sensors a device exposes and recognizes an application’s own activity labels using as little labelled target data as the deployment can afford. Progress has been pursued mainly through scale, but IMU signals admit no closed grammar the way language does—a change of device orientation, mounting, or on-device filtering can place a recording outside any corpus—so more pretraining data alone does not confer compatibility. A second, subtler obstacle is that the field’s standard remedy for open-set labels, projecting sensor features into a language embedding space, presumes that sensor similarity tracks linguistic similarity. It does not: *drinking coffee* and *drinking water* are near-identical motions but distant strings, whereas *sitting* and *sitting down*—a static posture and a dynamic transition—are nearly identical strings. We introduce HALO (**H**eterogeneity-**A**ware **L**anguage-conditioned **O**pen-set model)¹, which responds to each obstacle with a design choice rather than with scale. First, a tokenizer with deliberately *calibrated inductive bias*: co-located axes are grouped into sensors, each read on a fixed physical-frequency axis by a constant-Q filterbank over a native-rate transform, making the model rate-invariant and anti-aliased *by construction*, while sensor identity (device, modality, placement, gravity convention) is supplied as natural language so that channel count and ordering remain free. This is enough physical structure to survive unseen configurations, and few enough assumptions to compose heterogeneous corpora—in contrast to layout-locked encoders, which assume too much, and per-channel free-text models, whose conditioning can degenerate into memorising dataset provenance. Second, an *evidence engine* that avoids reconciling the two geometries at all: instead of projecting sensors into label-text space, it retrieves encoded exemplars from a memory by *sensor* similarity and accumulates evidence for whichever candidate labels

the deployment declares at run time. Representation pre-training is entirely activity-label-free; labels enter only as annotations on retrieved memory entries. Together these yield a single *label-efficiency* curve rather than a zero-shot-versus-fine-tuning dichotomy: the same frozen model recognises declared labels with no target data, improves as a handful of demonstrated examples are appended to memory without any gradient update, and remains available for conventional fine-tuning when labels are plentiful. We train on 12 public HAR datasets and evaluate on 7 held-out datasets under a leakage-free, subject-disjoint protocol that scores against each dataset’s own label strings, and package the model behind a small `predict()` interface, demonstrated in an on-device iOS app.

1 INTRODUCTION

A physiotherapist discharges a patient after knee surgery with six prescribed exercises, and wants to know whether they were actually done. The patient already owns a phone and a watch; the six exercises were chosen for this patient from a long tail of clinical options; and nobody—not the clinic, not the patient—is going to collect and annotate a training set. This is the ordinary case for inertial sensing, and it is not what the field builds for. Almost every deployed IMU system today is constructed the other way round: fix the sensor layout, fix the label set, collect a bespoke annotated corpus, train a classifier for it. That pipeline works, and it has to be paid for again from scratch for the next clinic, the next exercise list, and the next phone model.

Three properties of the situation above are what make it hard, and each is a property of the *deployment* rather than of the recognition problem. The label set is **open**—defined per patient, per workplace, per study, and containing activities no public corpus has ever recorded. The sensing is **heterogeneous**—whatever hardware the user happens to own, worn wherever they put it, sampled at whatever rate the firmware chooses. And the supervision budget is **small and fixed**—a clinician can demonstrate an exercise a handful of times inside one appointment, and cannot label five hundred repetitions. Human activity recognition (HAR) has no foundation

¹The source code and pretrained models for HALO will be released upon acceptance.

model that meets all three at once: one pretrained model a developer picks up, points at whatever sensors a device exposes, tells the application’s own activity names, and asks for only as much labelled target data as the deployment can actually afford. Section 3.1 argues these three demands from concrete deployments, and why they are not a wish-list but the minimum for the work to be reusable at all.

The field’s dominant answer has been scale. We argue that scale alone cannot deliver this, for two reasons that are specific to inertial sensing, and that these two reasons dictate two design choices rather than a bigger corpus.

(L1) IMU admits no closed grammar. Language and vision foundation models work because their substrate is, in a real sense, covered by a large corpus: text is generated by a grammar and a finite lexicon, and natural images obey a stable projective geometry, so enough data does approach the space of possible inputs. Inertial signals have no comparable closure. The same motion, recorded from the same body, becomes a different signal if the device is rotated, worn on a different limb, sampled at a different rate, low-pass filtered by an unknown firmware stage, or reported as gravity-inclusive total acceleration rather than the gravity-removed convention of a different operating system. These are not rare corner cases; they are what changes when an app ships on a second phone model. A larger corpus reduces the *probability* of surprise, but it cannot close the space, because the transformations are continuous and compose. Compatibility with an unseen acquisition configuration therefore has to be built in as *inductive bias*, not learned from coverage.

Inductive bias, though, is a quantity to be *calibrated*, not maximised. Assume too much—a fixed channel layout, a fixed sampling rate—and the encoder cannot read a device it was not built for. Assume too little—treat every scalar channel as an independent series carrying only a free-text name—and nothing prevents the model from using that text as a dataset identifier, since in a multi-corpus training set the description of a channel is very nearly a label for which corpus it came from. HALO’s tokenizer occupies the middle: it commits to the physics that is genuinely universal (co-located axes form a rigid sensor; motion energy lives at physical frequencies; gravity is a signed DC offset) and refuses to commit to anything that varies by deployment (rate, channel count, ordering, placement, which sensors exist at all).

(L2) Sensor similarity does not track linguistic similarity. The standard remedy for open-ended label sets is to project sensor features into a sentence embedding space and retrieve the nearest label

string [47, 57, 84]. This presumes that the two geometries agree. They do not, and they disagree in both directions. *Drinking coffee* and *drinking water* are the same wrist trajectory but only 0.56 apart in text; *sitting* and *sitting down*—a static posture and a dynamic transition, about as different as two IMU windows get—sit at 0.93. Across a set of 14 label pairs we selected for this contrast and measured with the same frozen sentence encoder the field uses (Section 3.4), 38 of the 49 cross-comparisons are *inverted*: a pair of physically distinct activities is scored more similar than a pair of physically identical ones. A single projection asked to reconcile these two geometries is mis-specified, and the easiest way for it to reduce training loss is to exploit whatever *does* predict the label within a corpus—subject, session, device—which is precisely the dataset memorisation that cross-dataset evaluation is meant to catch.

HALO. We present HALO (**H**eterogeneity-**A**ware **L**anguage-conditioned **O**pen-set model), which answers each obstacle with a design choice. Against L1, a tokenizer with calibrated inductive bias: co-located axes are grouped into *sensors*, each read on a fixed physical-frequency axis by a constant- Q filterbank computed from a native-rate transform, so the representation is rate-invariant and anti-aliased *by construction*—no resampling exists anywhere in the path—while per-token observability masks state which frequency bands a given rate and window could actually measure rather than fabricating them. Sensor identity (device, modality, placement, gravity convention) enters as natural language attached to the *sensor*, not to each scalar channel, so channel count and ordering are free while the conditioning stays too coarse to serve as a corpus fingerprint. Against L2, an *evidence engine* that declines to reconcile the two geometries at all: rather than projecting sensor features into label space, it retrieves encoded exemplars from a memory by *sensor* similarity and accumulates evidence for whichever candidate label strings the deployment declares at run time. Representation pretraining is entirely activity-label-free; labels enter only as annotations on retrieved memory entries.

(L3) The quantity that matters is label efficiency. These two choices are usually argued about under the banner of “zero-shot versus fine-tuning,” which we think is a false dichotomy and a bad way to measure the thing. A deployment does not choose between zero labels and a full annotation campaign; it has a budget, and the question is what that budget buys. HALO is therefore evaluated on a single *label-efficiency curve*: the same frozen model recognises declared labels with no target data at all, improves as a handful of demonstrated examples are appended to memory with no gradient

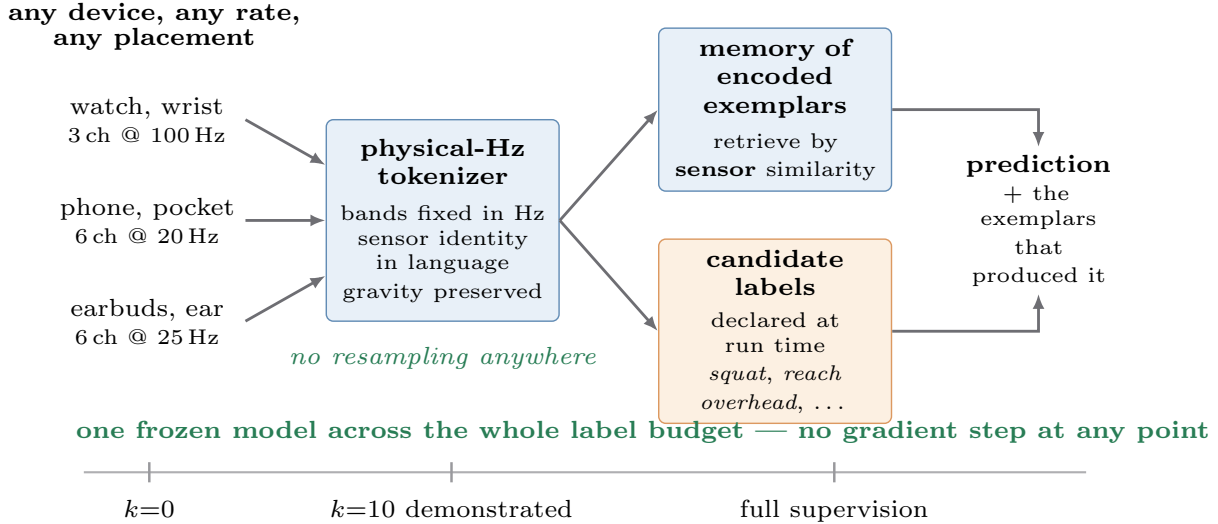


Figure 1: HALO absorbs acquisition heterogeneity *in the tokenizer*—a fixed physical-frequency axis, sensor identity supplied in language, gravity preserved—so that windows recorded on different devices are directly comparable as vectors. Recognition is then a retrieval in *sensor* space against a memory of encoded exemplars, with the label strings the deployment declares at run time used only to name and to enumerate. Because the memory is append-only, one frozen model spans the whole label budget, from no target data to full supervision, with no gradient update at any point.

update and nothing leaving the device, and remains available for conventional supervised fitting when labels are plentiful. Zero-shot is the left endpoint of that curve, not a separate claim.

We train on 12 public HAR datasets spanning 20 sensor streams and 93 activity labels, and evaluate on 7 held-out datasets under a leakage-free, subject-disjoint protocol that scores against each dataset’s own label strings.

We make the following contributions:

- We argue that inertial sensing has *no closed grammar*, so cross-configuration compatibility must come from calibrated inductive bias rather than from corpus coverage, and we realise that bias as a tokenizer that is rate-invariant and anti-aliased by construction, groups co-located axes into rigid sensors, and conditions on *per-sensor* rather than per-channel language—coarse enough to describe an unseen device, too coarse to memorise which corpus a window came from.
- We show, by direct measurement on our own 93-label vocabulary, that sensor similarity and linguistic similarity are not merely imperfectly correlated but frequently *inverted*, and we draw the architectural consequence: HALO retrieves in sensor space and votes in label space through an *evidence engine*, so the two geometries are never forced through

one projection. Representation pretraining uses no activity labels at all.

- We reframe the evaluation as a single *label-efficiency curve* spanning $k=0$ declared-label-only recognition, k demonstrated exemplars appended to memory without any gradient update, and full supervision—and we establish a leakage-free, subject-disjoint cross-dataset protocol that scores every model against each target dataset’s own label strings, with a dataset-identity probe that tests directly whether an encoder’s conditioning has degenerated into corpus memorisation.
- We package the model behind a small `predict()` interface and demonstrate it in an on-device iOS application. We will release the pretrained checkpoint and code as a reusable foundation for mobile HAR.

2 RELATED WORK

HALO sits at the intersection of four literatures, and in each of them some part of what we do already exists. We name the closest work in each and state what is left.

Cross-dataset and foundation models for IMU sensing. Deep HAR has moved from single-dataset CNN–LSTM models [54] through methods that attack one axis of heterogeneity at a time—contrastive fusion for small-data multimodal HAR [55], physics-informed augmentation for cross-user and cross-device transfer [79],

federated modelling of inter-user structure [56]—to self-supervised encoders intended to transfer: LiMUBERT [80] pretrains by span-masked reconstruction at 20 Hz, CrossHAR [27] combines masked reconstruction with contrastive learning, and ssl-wearables [86] scales multi-task self-supervision to roughly 700k person-days of wrist accelerometry. A second wave targets foundation-scale wearable modelling: LSM [50] establishes the label-efficiency argument (up to $8\times$ fewer labelled examples), LSM-2 [82] learns directly from incomplete recordings, RelCon [81] learns a motion-similarity metric over $\sim 10^9$ windows, and SensorLM [93] pairs sensor data with hierarchical captions and reports zero-shot, few-shot, and cross-modal retrieval together. General time-series models [23, 52] and tabular ones [90] trade domain-specific bias for breadth. A recent survey maps this space [7], and Inertia-1 [83] runs a controlled lifecycle study over modality, placement, rate, and window length. *What is left*: all of these treat sampling rate as a preprocessing decision to be normalised away. Our argument is that this is where the information loss happens, and Section 6.3 measures it rather than asserting it.

Sampling-rate and resolution invariance. This is the literature closest to our L1 mechanism, and it is not in HAR. Rate-invariant autoencoding [35] learns representations quotiented by temporal warping; FlowState [24] builds a sampling-rate-*equivariant* forecasting model on a state-space encoder with a functional-basis decoder; universal spectral tokenization [63] ingests native, irregular wavelength grids without resampling and explicitly proposes the same idea for multiresolution time series. Within IMU sensing, MoPFormer [88] tokenizes into learned motion primitives to make datasets comparable, but resamples every dataset to 100 Hz before quantising, so its codebook is defined over a fixed-rate grid. *What is left*: FlowState conditions on the rate—it is equivariant, adapting its computation—whereas HALO’s features are *anchored* to physical Hz, so a band denotes the same physical frequency for every device and no conditioning is required to compare across them; and the astronomical setting of [63] has no gravity, no Nyquist guard tied to a wearable’s firmware, and no anti-aliasing consequence to demonstrate. To our knowledge no prior HAR system claims rate-invariance by construction, and none measures what resampling costs on held-out data—which is the experiment we run, and which we regard as the load-bearing evidence for L1 rather than the mechanism itself.

Language as an interface to sensors. Two distinct uses of language must be separated. On the *output* side, aligning a modality to sentence embeddings for open-vocabulary recognition descends from DeVISE [20],

ConSE [53], and CLIP [57]; ImageBind [22] extends it to IMU among other modalities, and LanHAR [84] and LLaSA [28] bring it to HAR. On the *input* side—describing the sensor rather than the activity—the lane is crowded and recently so. GOAT [47] supervises with natural language over both activity labels and device on-body positions. ZeroHAR [15] conditions on textual sensor context—sensor type, Cartesian axis, body position—and aligns the motion latent to it before aligning to activity text. CHARM and its successor [17, 18] feed channel-level textual descriptions into a channel-order-equivariant transformer, and their ablation finds that conditioning on channel identity via text is one of the largest performance drivers. SensorLLM [40] generates per-channel descriptions to teach an LLM the sensor layout; oneHAR [76] handles variable combinations of sensor positions with LLM assistance; UniMTS [91] takes the complementary route of encoding placement structurally as a joint graph with rotation-invariant augmentation. *What is left, honestly*: “sensor identity as natural language” is not ours to claim. Our differences are specific and narrower than the general idea. First, granularity: ZeroHAR conditions on a fixed, closed set of categorical attributes, whereas ours is free text and therefore composes to descriptions never seen; the channel-description line [17, 18, 40] is free text but attaches it per-channel, which is exactly the configuration whose corpus-identifiability we argue against and test in Table 9. Second, what the conditioning feeds: in all of the above it feeds a parametric head, whereas ours feeds a non-parametric memory. Third, invariance versus conditioning: UniMTS’s rotation invariance discards orientation, and with it gravity and posture, which conditioning preserves.

Retrieval, memory, and few-shot recognition. Deciding by comparison to stored examples is old in this field and new in its current form. The ubicomp lineage recognises gestures by template: uWave [41] needs a single example per gesture and matches by dynamic time warping; Junker et al. [30] preselect candidate segments by similarity search before classifying, which is structurally the retrieve-then-decide shape we use; WarpingLCSS [51] does robust online template spotting from crowdsourced annotations. The modern lineage is episodic and parametric—matching networks [75], prototypical networks [68]—and, in language and vision, non-parametric memory attached to a frozen model [9, 29, 33, 38, 49]. Memory-modular classification [32] states our Phase-B property as its contribution: generalise to new classes by replacing memory

contents, without retraining. In sensing specifically, RAG-HAR [67] performs training-free retrieval over signal descriptors and lets an LLM decide, recognising unseen activities; ZARA [39] retrieves multi-sensor evidence into an LLM agent, under the same “evidence-grounded” name we use for a different mechanism; and in-context time-series classifiers [19, 36] take a labelled support set and a query in one forward pass with no gradient step, collapsing zero-shot and few-shot into a single inference mode. *What is left:* not “we retrieve.” The specific claim is that retrieval is performed in a space where similarity is *sensor* similarity between encoded exemplar *patches*, produced by an encoder that was conditioned on acquisition configuration—which is what lets a wrist query at an unusual rate retrieve comparable evidence, and which descriptor-space [67], LLM-agent [39], and CLIP-space [93] retrieval cannot do. To that we add calibrated rejection [21, 62], which none of the retrieval-based HAR systems above provide. Section 6.2 reports the control that decides whether this distinction is real: the same memory mechanism on a competitor’s frozen encoder.

Self-supervised objectives. Phase A composes established pieces: contrastive [14, 72] and supervised contrastive [34] learning as controls; non-contrastive objectives that avoid declaring in-batch negatives [5, 87]; masked modelling from language and vision [16, 26]; and latent-target prediction against a momentum teacher [2, 3, 25]. Our time–frequency term is inspired by TF-C [92] but is not faithful to it, and we say so where it is described. The contribution here is not a new objective but the choice to remove activity labels from all of them, and the cross-placement positives that a corpus with verified simultaneous multi-device recordings makes available.

2.1 The Combination, and Why It Is Not a List

Table 1 summarises which axes each of the closest systems explicitly addresses. No prior system addresses all of them, which is true but is also the least interesting thing to say about it: “we combine more components” is not a contribution, and a reviewer is right to be unmoved by a filled-in row. Variable channel counts have known solutions; variable sampling rates have known solutions; language as a bridge to an open label vocabulary has known solutions. We claim none of those as novel.

We make a different kind of claim: that physical-Hz *anchoring* is what makes a single, flat, cross-device memory well-posed in the first place—an *enabling* result rather than an assembly. A non-parametric memory shared across deployments requires that a query encoded on

one device and an exemplar encoded on another be comparable *as vectors*, with no per-configuration index and no conditioning step to reconcile them. Resampling does not provide this (it discards content the two devices do not share), and rate-equivariance does not provide it either (it relates the two by a learned transformation rather than making them the same quantity). This is why the existing retrieval-based sensing systems retrieve somewhere else—in hand-designed descriptor space [67], through an LLM agent [39], or in CLIP space [93]—and why none of them needs the property we are claiming. Section 6.3 tests it directly, using recordings in which two devices captured the same person performing the same activity at the same instant, so that configuration is the only variable (Table 7).

Once that claim is in view, the remaining choices stop looking like options. They are *mutually entailed*—each one is what makes another valid—so the set is a consequence of L1 and L2 rather than a collection of independently attractive features. And because “it is just a combination” is an empirical objection, we answer it empirically as well: the conventional assembly of the same ingredients—resample to a common rate, attach free text per channel, put retrieval on top—is run as a rival arm in Table 8, not merely argued against.

Rate anchoring is a precondition for the memory, not a parallel feature. A non-parametric memory only works if a query and a stored exemplar recorded on different devices are comparable *as vectors*. If the encoder is *rate-equivariant*—told the rate and adapting—then two windows at different rates are related by a learned transformation, and the memory must either be indexed per configuration or trust that transformation to be exact. Because HALO’s band k denotes the same physical frequency for every device, one flat memory is well-posed. L1 exists in the form it does because of Phase B.

The granularity of the language conditioning is forced by the retrieval, too. Per-channel free text [17, 18, 40] is the natural choice if the conditioning feeds a classifier: it is maximally expressive and the head can ignore what it does not need. It is the wrong choice if the conditioning feeds a memory, because a description that is nearly unique to a corpus makes *same-corpus* neighbours the nearest ones, which is precisely the shortcut cross-dataset evaluation exists to catch. Attaching identity to the *sensor* keeps the description expressive enough to name an unseen device and too coarse to name a dataset. That is why Table 9 is in the paper.

Preserving gravity is forced by the geometry finding. Rotation invariance [91] is the standard answer

System	acquisition		orient./	label	k-curve
	rate	placement	gravity	interface	(no grad.)
LiMU-BERT [80], CrossHAR [27]					✓
harnet [86], LSM [50, 82]					
NormWear [43]		◦		✓	
UniMTS [91]		✓	◦	✓	
ZeroHAR [15]		◦		✓	
GOAT [47], oneHAR [76]		✓		✓	
CHARM [17, 18]		✓			
MoPFormer [88]		◦			
FlowState [24]	◦				
RAG-HAR [67], ZARA [39]				✓	✓
SensorLM [93]				✓	✓
HALO	✓	✓	✓	✓	✓

Table 1: Which axes each system *explicitly claims* to address. ✓ = addressed in the form we argue is needed; ◦ = addressed in a weaker or lossy form (rate *equivariance* rather than physical-Hz anchoring; *categorical* rather than free-text placement; rotation *invariance*, which discards gravity and therefore posture); blank = not claimed by that work. A blank is not a criticism—these are different papers solving different problems—and the table reports stated scope, not measured performance. The point last column marks works that report performance across a range of label budgets; HALO’s mark there differs in kind rather than degree, in that one frozen model traverses every budget with no gradient update at any point, whereas the other marked rows fit something at each budget. The point of the table is not the bottom row’s length; it is the dependency structure described below.

to unknown mounting, and it is lossy in one specific way: it discards the gravity direction, and with it posture. Section 3.4 shows that the label pairs a text projection cannot separate are exactly the posture-versus- transition pairs (*sitting/sitting down*, *lying/standing*). A system that gives up on the text geometry *must* therefore keep posture in the signal, so conditioning rather than invariance is not a preference here but a requirement.

Label-free pretraining is forced by the same shortcut argument. Any closed-vocabulary objective over a multi-corpus training set rewards inferring the corpus and predicting its marginal. That is survivable for a parametric head, which can be refitted per deployment; it is not survivable for a frozen encoder feeding a memory, because the shortcut is baked into the stored vectors.

So the four choices form a chain: removing any one of them invalidates another. That is a different and more falsifiable claim than “we do all of these,” and Section 6.3 is built to test it—each ablation in Table 8 removes one link and asks whether the rest still stand.

A note on scale. We do not think the scale-first programme is naive, and we do not claim it fails on

its own terms. On population-scale health outcomes, models trained on tens of millions of hours [50, 86, 93] are the right instrument and we would not beat them. The disagreement is about which axis a corpus can close. Scale reduces surprise along the axes a corpus *samples*—subjects, activities, sessions. It cannot close the axes that are *generated at deployment time* by a continuous, composable chain of mounting, placement, filtering, and convention choices, because those are not drawn from the corpus distribution at all. Our position is not “less data, cleverer model”; it is that these two axes call for different remedies, and that the second one has been under-served because it is invisible to the benchmark protocols the field uses.

3 MOTIVATION

Section 3.1 argues from concrete deployments why open-set labels, sensing heterogeneity, and a small supervision budget are the binding constraints, rather than features a paper might optionally offer. Section 3.2 then characterises the heterogeneity technically. Section 3.3 argues that it cannot be closed by data—inertial signals admit no closed grammar—so compatibility must be supplied as inductive bias, and that the bias must be *calibrated* rather than maximised. Section 3.4 shows by direct measurement that the field’s standard mechanism for open-ended labels rests on a false premise. Section 3.5 turns these into design requirements.

3.1 Why This Problem: Three Deployments and What They Demand

That IMU-based activity recognition is useful needs no argument. What needs an argument is why it should be *open-set*, *heterogeneity-aware*, and *label-efficient*—three qualifiers that add real difficulty and that a system could reasonably decline to take on. Our answer is that in the deployments that motivate the work, all three are forced simultaneously, and dropping any one of them removes the application rather than simplifying it.

Rehabilitation adherence. A physiotherapist prescribes a set of exercises after surgery and needs to know whether they were performed, and performed correctly. The exercise set is chosen *per patient* from a long clinical tail—a specific loaded knee extension, a specific scapular retraction—so no fixed output vocabulary covers it. The sensor is the patient’s own phone or watch, worn wherever they choose. The available supervision is whatever the clinician can demonstrate inside a twenty-minute appointment.

Deployment	open-set labels	heterogeneous sensing	small label budget
Rehabilitation adherence	✓	✓	✓
Occupational strain	✓	✓	✓
Free-living monitoring	✓	✓	✓
<i>Public HAR benchmark</i>	—	—	—

Table 2: The three qualifiers are not independent features to be offered separately. Every deployment that motivates this work demands all three at once, because all three are properties of the deployment rather than of the recognition task. A public benchmark demands none of them—which is why a model can score well on one and still be unusable in any of the rows above.

Occupational strain. A safety team wants to count a specific risky motion in one facility: reaching under a chassis, lifting from below knee height. The vocabulary belongs to that facility and its risk assessment. The hardware is a fleet accumulated over years, mixing phone models and wristbands. Collecting a labelled corpus would require instrumenting the floor with video and obtaining consent from every worker in frame—which is precisely the thing the deployment exists to avoid.

Free-living behavioural monitoring. A cohort study measures posture, sedentary time, and sleep over weeks. Devices vary across participants and firmware revisions vary within a participant; the categories are the study’s own, defined by its protocol rather than by a benchmark.

Table 2 makes the pattern explicit, and also makes clear why benchmark performance has been a poor guide: a fixed-vocabulary, fixed-layout, fully-supervised benchmark demands none of the three, so a model can lead it and still be inapplicable to every row above. We now take each qualifier in turn, because each rests on a different argument.

Why open-set is not optional. Two reasons, neither of which is solved by more data. First, *the label set is not the model’s to choose*. A prescribed rehabilitation exercise is defined per patient; a task vocabulary is defined per facility. These are not rare classes that a larger corpus would eventually cover—they are defined after training, by someone who is not us. A model with a fixed output layer is structurally unable to express them. Second, *the annotation channel for inertial data is uniquely poor*. Images and text can be labelled from the artifact itself, which is why annotation crowdsources and why those corpora are enormous. Nobody can label an accelerometer trace by looking at it. Labels must therefore come from synchronised video, scripted in-laboratory protocols, or self-report—all expensive, and all biasing

the resulting data toward laboratory-like conditions. This is the mechanical reason HAR corpora are small and scripted while image corpora are large and natural, and it will not be fixed by effort. A system that requires a labelled target set per deployment inherits that cost every time.

Why heterogeneity-aware is not optional. The cost of sensing heterogeneity is usually described as a one-time integration expense. It is not: it *recurs*, per device model, per placement, per firmware revision, for as long as the product ships. A vendor covering fifty device-by-placement combinations either maintains fifty models—fifty training runs, fifty validation cycles, fifty entries in a model registry—or ships one model built for the lowest common denominator. In practice the second happens, and it is a large part of why deployed phone activity recognition is still limited to a handful of coarse ambulatory classes. It is also why there is no “download the model and point it at your sensor” moment for IMU as there is for vision: the model cannot read your sensor. Notably, this is not only our reading. UniMTS [91], the leading zero-shot motion model, ablates its own components and attributes substantially more of its performance to acquisition-configuration handling—rotation-invariant augmentation and placement encoding—than to its label-text handling. The field names sensor heterogeneity and poor activity text as its barriers; it does not name label semantics.

Why the label budget is the right axis. Given the two arguments above, the useful question is not whether a system is “zero-shot” but how much target supervision it needs to become useful, and what that supervision costs to obtain. Zero labelled examples is the interesting left endpoint because it is the only point reachable without any target data collection at all. But a clinician demonstrating an exercise five times is a realistic and cheap form of supervision, and a system that can consume it *without a training cycle* is categorically easier to deploy than one that requires fine-tuning: nothing needs to be retrained, nothing needs re-validating, and the demonstration never has to leave the device. That last property is a legal precondition in the clinical and workplace settings above, not a convenience.

What this buys, and why it should interest a systems audience. The result of meeting all three is a difference in *capability*, not a difference in accuracy: one encoder serving many deployments instead of one model per configuration; new activities and new devices added by appending to a memory rather than by re-training, so that a model already validated does not have to be validated again; enrollment performed on-device; and predictions that can be inspected, because a

retrieval-based decision can name the stored examples that produced it. In a regulated or clinical setting the last two are worth more than a few points of macro-F1.

3.2 Sensing Heterogeneity Across Devices and Deployments

Wearable and mobile devices differ widely in their sensing configurations, and the differences are not cosmetic. Our training corpus alone (Table 3, Figure 2) spans sampling rates from 20 to 100 Hz, accelerometer-only and accelerometer-plus-gyroscope suites, and sensor placements at the waist, trouser pocket, wrist, lower back, ear, and head—the last two from smart glasses and wireless earbuds, devices that did not exist as HAR platforms when most public benchmarks were collected.

Sampling rate. At 20 Hz a 6-second window is 120 samples; at 100 Hz it is 600. Any encoder whose positional code is tied to a timestep index conflates physical time with sequence length, so the same motion at two rates presents as two different structures. The usual fix—resample everything to a common rate—is worse than it looks: downsampling to the corpus minimum aliases the high-frequency content that distinguishes, say, brushing teeth from waving, and upsampling fabricates samples that were never measured.

Channel set, placement, and orientation. Configurations range from a single 3-axis wrist accelerometer to six simultaneous placements on one body. A fixed-width input layer cannot accommodate this without padding artifacts or per-dataset input heads. Worse, the *meaning* of a channel depends on where and how the device is worn: a wrist accelerometer’s x -axis and a hip accelerometer’s x -axis are not the same physical quantity, and neither is stable if the user pockets the phone upside down.

Conventions the data does not announce. Two accelerometer streams that look identical may differ in gravity convention (Android reports total acceleration; iOS `userAcceleration` removes gravity), in units, or in an unadvertised on-device low-pass filter. In assembling our corpus we found several public datasets whose distributed form had already been transformed—one is per-axis min-max normalised, destroying physical scale; another ships gravity-removed body acceleration under a name suggesting total acceleration. A model that silently assumes a convention will be wrong on data it has every reason to think it understands.

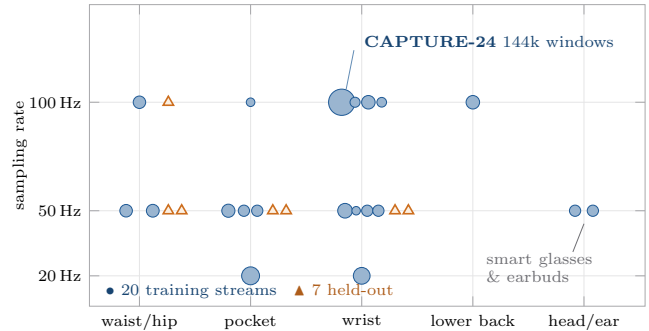


Figure 2: The corpus does not cover a grid; it covers a scatter. Each blue disc is one of the 20 sensor streams HALO trains on, positioned by body placement and native sampling rate, with area proportional to the number of 6 s windows it contributes; orange triangles are the 7 held-out evaluation datasets. Rates span a $5\times$ range and placements include two—smart glasses and wireless earbuds—that no established HAR benchmark contains. Held-out cells are not simply new subjects: they combine placements and rates in proportions the training corpus never sees. Absorbing this by resampling to a common rate would discard the content above 10 Hz that only the 50/100 Hz streams can resolve.

3.3 Inertial Signals Admit No Closed Grammar

The heterogeneity above is usually treated as an engineering nuisance to be absorbed by pretraining on more of it. We think that is the wrong diagnosis, and the difference matters because it changes what the solution has to be.

Foundation models in language and vision work in large part because their input space is effectively *closed* by a corpus. Text is produced by a grammar over a finite lexicon; enough text does approach the distribution of sentences a model will meet. Natural images obey a stable projective geometry and a stable statistics of surfaces and light. In both cases the substrate has structure that a sufficiently large sample covers.

Inertial signals have no such closure. The map from a motion to a recorded signal is composed at deployment time from a chain of choices—mounting orientation (a continuous $SO(3)$ nuisance), body placement, sampling rate, filter chain, gravity convention, sensor suite—and the chain composes. There is no finite enumeration to cover, and the free parameters are continuous. A corpus $10\times$ larger reduces the *probability* that a new deployment is far from anything seen, but the tail does not close, and the transformations that generate it are cheap

to encounter: shipping on a second phone model, or a firmware update, is enough.

If compatibility cannot come from coverage, it must come from *inductive bias*: the architecture, not the data, must make the representation invariant to the nuisance axes and explicitly aware of the ones that carry signal. This is the standard move—but for inertial sensing it has a failure mode on *both* sides, and the interesting design question is where to sit between them.

Too much bias. Layout-locked encoders fix a channel order, a channel count, and a rate. They are the most sample-efficient thing to train and cannot read a device they were not built for. Orientation-*invariant* designs (e.g. training against uniformly sampled $SO(3)$ rotations) are a subtler version of the same error: invariance is lossy. Discarding orientation discards gravity, and with it the distinction between lying down and standing still, which is exactly the kind of activity a posture or sleep application cares about.

Too little bias. The opposite extreme treats every scalar channel as an independent univariate series identified only by a free-text name. This is maximally flexible, and it creates a shortcut. In a multi-corpus training set, the string describing a channel is nearly a sufficient statistic for *which corpus* the window came from; so is the sampling rate, and so is the channel count. A model rewarded for predicting a label within a corpus can satisfy much of that objective by first inferring the corpus and then predicting its marginal—and cross-dataset evaluation is precisely where that strategy stops working. We therefore treat corpus identifiability as a measurable failure and probe for it directly (Table 9).

A third position, recently explored outside HAR, is to make the model *equivariant* to resolution—to condition its computation on the sampling rate so that it adapts, as in sampling-rate-equivariant forecasting [24] or grid-agnostic spectral tokenization [63]. That is strictly better than resampling, but it is not the same as what we want. An equivariant model still has to be *told* the rate, and its features at two rates are related by a learned transformation rather than being the same quantity. We want the stronger property: that a feature *denotes* a physical frequency, so a band computed from a 20 Hz phone and a band computed from a 100 Hz research IMU are directly comparable without any conditioning at all—which is what makes them comparable in a shared memory. Within IMU sensing this property is, as far as we know, unclaimed: the closest tokenizer, MoPFormer [88], resamples every dataset to 100 Hz before quantising into motion primitives, so its codebook is defined over a fixed-rate grid.

The calibrated position, which HALO adopts, is to commit to the physics that does not vary and to nothing that does. Three things do not vary: co-located axes on one rigid device share a coordinate frame and must be rotated together, not independently; human motion energy lives at physical frequencies, so a band means the same thing at every sampling rate; and the quasi-static component of an accelerometer is gravity, so it carries posture and must be preserved as a signed quantity rather than normalised away. Everything else—rate, channel count, ordering, placement, which sensors exist—is deferred to run time and described in language, at the granularity of the *sensor* rather than the channel, so that the description is expressive enough to name an unseen device but too coarse to fingerprint a corpus.

3.4 Sensor Similarity Does Not Track Linguistic Similarity

The second obstacle concerns the output side. Because deployments name their own activities, a reusable model cannot have a fixed output layer, and the field’s answer has been to embed label strings with a frozen sentence encoder and align sensor features to that space, so that recognition becomes nearest-neighbour retrieval over an arbitrary label library [47, 57, 84, 91]. The mechanism is elegant and its premise is rarely stated: that proximity in sentence-embedding space is a usable proxy for proximity in motion space.

It is not. Figure 3 measures this on our own vocabulary. We selected 14 label pairs—seven where the two names denote the same physical motion, seven where they denote different motions—and computed the cosine similarity of their embeddings under all-MiniLM-L6-v2, the encoder these systems use. If the premise held, the same-motion pairs would sit to the right of the different-motion pairs. The ordering is reversed: the different-motion pairs average 0.77 and the same-motion pairs 0.60, and 38 of the 49 cross-comparisons are individually inverted. *Drinking coffee* and *drinking water* are the same wrist trajectory at 0.56; *sitting* and *sitting down*, a static posture versus a dynamic transition, are at 0.93.

The failure is structural rather than a deficiency of one sentence encoder. Natural language encodes activities by their *object and purpose*—what is being drunk, what is being cut—while an IMU sees only the *kinematics of the limb*, which is invariant to the object. Conversely, morphological relatives (*sit/sit down*, *upstairs/downstairs*) share nearly all of their surface form while denoting opposite dynamics. No amount of alignment training repairs a target geometry that is wrong; a model forced to

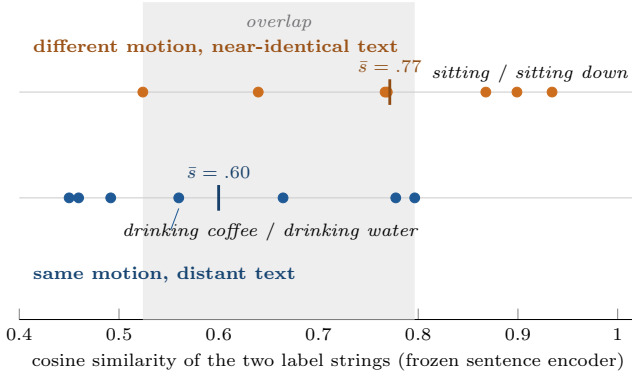


Figure 3: Sensor similarity and linguistic similarity are inverted, not merely noisy. Each dot is one pair of activity names from our label vocabulary, placed at the cosine similarity of their frozen sentence embeddings (all-MiniLM-L6-v2, the encoder language-aligned HAR models use). Pairs in the *top* lane denote physically distinct activities; pairs in the *bottom* lane denote the same physical motion. If text similarity tracked sensor similarity the lanes would separate, with the bottom lane on the right. They separate the wrong way: the mean of the physically *distinct* pairs (0.77) exceeds the mean of the physically *identical* pairs (0.60), and 38 of the 49 cross-lane comparisons are inverted. The motion grouping is our judgement; the similarities are measured. A single projection from sensor space into this text geometry is therefore mis-specified—which is why HALO retrieves in sensor space and votes in label space (Section 5.3).

fit it will minimise its loss by other means, and within a multi-dataset corpus the cheapest other means is dataset identity.

The architectural consequence is to stop trying. HALO keeps language as the *interface* for naming activities—that part is necessary, and cheap—but does not use it as the *metric space* in which recognition happens. Retrieval is performed in sensor space, where similarity means kinematic similarity; label text enters only afterwards, to name the retrieved neighbours and to enumerate what the deployment is willing to predict (Section 5.3).

3.5 Design Requirements

The three claims yield four requirements.

R1 (from L1). The encoder must accept arbitrary sampling rates, channel counts, sensor suites, and placements with one weight set and no architectural change, absorbing them through construction rather than

through resampling or per-dataset heads—and must signal what a given configuration could not observe instead of silently fabricating it.

R2 (from L1). The conditioning that carries acquisition context must generalise compositionally to unseen configurations, and must be coarse enough that it cannot serve as a corpus identifier. This is a testable property, not an assertion.

R3 (from L2). Recognition over a run-time-declared label set must not require a single projection reconciling sensor geometry with sentence geometry. Similarity must be computed where it is meaningful, and label text used only where it is meaningful.

R4 (from L3). The model must place no floor on how much target supervision is required, and adding a small amount must be possible without a gradient update, without retraining, and without data leaving the device—so that $k=0$, $k=10$, and full supervision are points on one curve rather than three different systems.

HALO meets R1–R2 with the physical-frequency filterbank tokenizer and factored per-sensor language conditioning (Section 5.1), R3 with the evidence engine (Section 5.3), and R4 because the evidence engine’s memory is append-only (Section 5.4).

4 SYSTEM OVERVIEW

HALO is organised as two phases with a hard boundary between them (Figure 4). **Phase A** learns a representation of inertial signals using no activity labels whatsoever. **Phase B** turns that frozen representation into a recogniser for whatever label set a deployment declares, using labels only as annotations on stored exemplars. The boundary is not an implementation convenience; it is where the argument of Section 3 lands. L1 says compatibility must be structural, so it belongs in the encoder, before any label is involved. L2 says the label decision must not be a projection into text space, so it belongs after the encoder, in a component that can be swapped per deployment.

Phase A: label-free representation pretraining (Section 5.1, 5.2). A window of raw, native-rate samples—at whatever rate the device runs, with whatever subset of accelerometer and gyroscope axes it exposes—is grouped into sensors and passed through a physical-frequency constant- Q filterbank, producing one token per sensor per patch on a frequency axis that is the same at every sampling rate. Each token is conditioned on natural language describing the *sensor* it came from, and on per-token masks declaring which frequency bands that rate and window length could actually observe. A dual-branch transformer with rotary positions

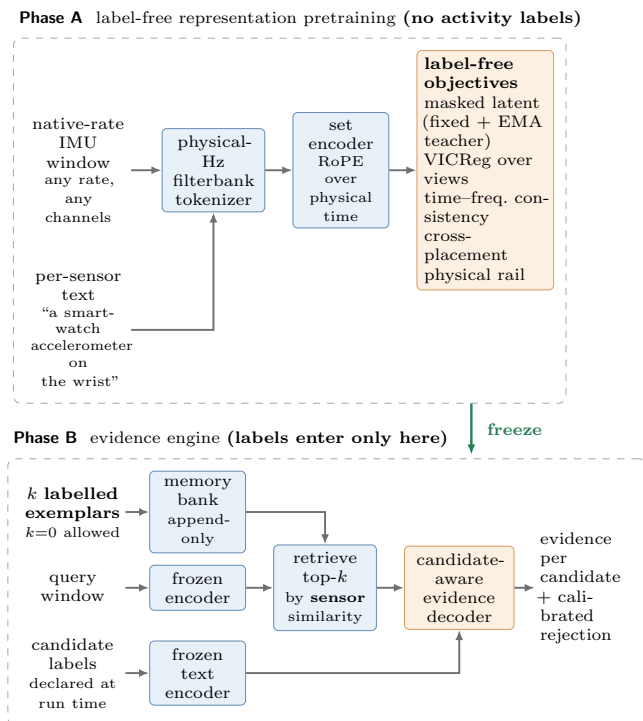


Figure 4: HALO’s two phases. Phase A trains the tokenizer and encoder with objectives that never see an activity label; heterogeneity is absorbed structurally (physical-Hz filterbank, physical-time positions, per-sensor language identity). The encoder is then frozen. Phase B encodes labelled exemplars into an append-only memory, retrieves neighbours of a query by *sensor* similarity, and lets a candidate-aware decoder accumulate evidence for the label strings the deployment declares at run time. Adding a new activity, or a new device, is an append to memory—no gradient step, no retraining.

over *physical time* then mixes tokens across time and across sensors. Training combines masked latent prediction against both a fixed physical target and a slow EMA teacher, a VICReg objective over independently augmented views, a time–frequency consistency term, a low-weight consistency term over *verified simultaneous* recordings of the same physical event at different body placements, and a small self-derived physical grounding rail. None of these uses an activity label.

Phase B: the evidence engine (Section 5.3). The encoder is frozen. Labelled windows—from the training corpus, from a target deployment, or both—are encoded once and stored in a memory table together with their label, subject, acquisition configuration, sensor group, and

patch timing. At inference, the query window is encoded and its patches retrieve neighbours from memory by similarity in *sensor* space. A decoder then contextualises query and evidence tokens together and accumulates non-negative evidence for each label string in the run-time candidate set, with a separately estimated rejection confidence derived from retrieval density, patch and sensor agreement, vote entropy, and the leading margin. Label text is used to name evidence and to enumerate candidates; it is never the space in which similarity is computed.

One curve, not two regimes. Because memory is append-only, the amount of target supervision is a dial rather than a mode. With an empty target memory the system still runs: evidence comes from corpus exemplars and the candidate strings alone. Appending k demonstrated windows per class costs one forward pass each and no gradient step, so it is feasible on a phone and the data need never leave it. When labels are plentiful the frozen representation also supports conventional supervised fitting. Section 6 therefore reports a single label-efficiency curve over k rather than separate zero-shot and fine-tuned tables.

Scale. The encoder is 7.17M trainable parameters ($d=256$, 6 layers; 6.32M in the transformer, 0.82M in text fusion, 25K in the filterbank), plus one frozen all-MiniLM-L6-v2 sentence encoder (22.7M) shared by sensor conditioning and label text. Phase-A objective heads add 0.62M and are discarded at inference. The Phase-B decoder is a small set transformer over retrieved evidence. **[TO BE COMPLETED]** *On-device latency, peak memory, and the deployed parameter count once the encoder is compiled for Core ML (Section 6.5).*

Deployment. A deployment is specified by configuration, not code: a per-sensor description string and a list of candidate activity names. Both are embedded once and cached, so run-time cost is one encoder pass plus a retrieval (Section 5.4).

5 DESIGN OF HALO

This section specifies the three components introduced in Section 4: the tokenizer that supplies HALO’s calibrated inductive bias (Section 5.1), the label-free objectives that train it (Section 5.2), and the evidence engine that turns the frozen representation into a recogniser over a run-time label set (Section 5.3). Everything described here is implemented; where a design choice has not yet been isolated by an experiment we say so and name the ablation that will isolate it.

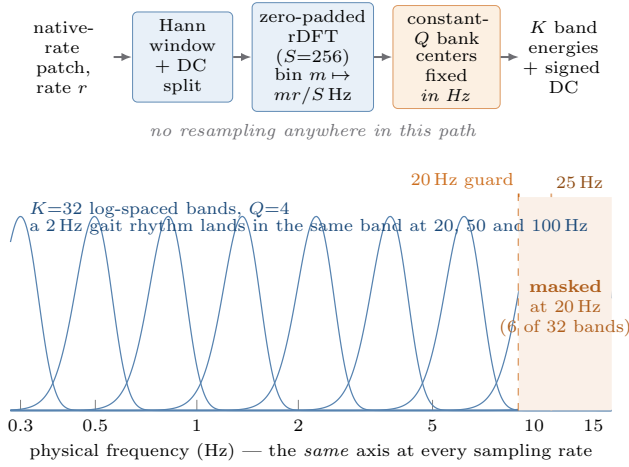


Figure 5: The physical-frequency constant- Q filterbank. Each native-rate patch is Hann-windowed, split into a signed quasi-DC component and an AC residual, transformed by a zero-padded real DFT whose bin m maps to the physical frequency mr/S , and read out by a bank of $K=32$ log-spaced Gaussian bands whose centers are fixed *in Hz*. Because the bin $\rightarrow$ Hz map absorbs the rate, the same band means the same physical frequency at 20, 50, and 100 Hz, and no interpolation exists anywhere in the path. Bands whose support crosses a rate’s Nyquist guard are *masked*, not fabricated: at 20 Hz six of the 32 bands are unobservable and are flagged as such to the encoder.

5.1 The Tokenizer: Calibrated Inductive Bias

5.1.1 Sensors, not channels. HALO’s first commitment is that the unit of description is the *sensor*, not the scalar channel. Three co-located accelerometer axes on one device share a coordinate frame: they rotate together, they experience the same gravity vector, and no physically realisable transformation moves one without the others. Treating them as three independent series discards this and, as Section 3.3 argued, opens the door to per-channel text acting as a corpus fingerprint. HALO therefore models a stream as a *set of sensors*, each an accelerometer or gyroscope triad with its own device, placement, and gravity convention. This grouping is what makes the augmentation physics correct as well: an $SO(3)$ rotation is applied jointly to an accelerometer triad and its co-located gyroscope, never to a single axis, so every augmented sample corresponds to a device orientation that could actually occur.

Accelerometer streams are gravity-aware. The quasi-static component below the analysis band is split off as a *signed* per-axis DC feature rather than removed, so posture survives: lying and standing differ in the DC term even when their AC content is indistinguishable. Because gravity conventions differ across platforms, the convention is declared in the sensor’s text (below) and a gravity-removal augmentation is applied with low probability so the encoder sees both conventions for the same underlying motion.

5.1.2 A physical-frequency constant- Q filterbank. Each patch of each channel is turned into one token entirely in the physical-frequency domain (Figure 5). The patch is Hann-windowed and its DC component removed, then transformed by a zero-padded real DFT of fixed length $S=256$. The zero-padding is what makes this rate-agnostic: for a patch sampled at rate r , DFT bin m corresponds to the physical frequency $\phi[m] = mr/S$, so the *same* matrix of bin indices maps to different Hz grids at different rates, and the readout can be defined once in Hz. That readout is a constant- Q Gaussian filterbank [10, 11] with $K=32$ centers log-spaced over $[0.3, 15]$ Hz and $\sigma_k = c_k/2Q$ at $Q=4$. The band limits are physically motivated rather than tuned: gait cadence occupies 0.5–3 Hz, limb and hand motion contribute harmonics to roughly 10–12 Hz, and content below 0.3 Hz is gravity, which the signed DC feature already carries. Band energies are window-normalised so they do not depend on patch length, log-compressed, and per-band standardised with statistics calibrated once before training.

Figure 6 shows this working on real data rather than asserting it. One 1.5 s wrist patch of walking is resampled to 100, 50, and 20 Hz and passed through the tokenizer at its own rate. As sample arrays the three are entirely different objects—150, 75, and 30 numbers—but the band energies they produce agree to cosine 0.997 and 0.996 over the bands all three can observe, and the six bands the 20 Hz stream cannot legally measure are masked rather than filled in.

The consequence is that rate-invariance is a property of the construction, not of a learned approximation: a 2 Hz gait rhythm deposits its energy in the same band whether the device sampled at 20 or 100 Hz, and because no interpolation exists in the path there is no aliasing to introduce and no fabricated sample to explain away. Note the distinction from *rate-equivariance* [24]: the model is never told the rate in order to compare two windows, because band k denotes the same physical frequency in both. That is what allows exemplars from different devices to occupy one memory (Section 5.3) without a

per-configuration index. Zero-padding beyond the true patch length is frequency-domain interpolation only and does not alter band energies; we verified that $S=256$ and $S=512$ give band-energy cosine 1.000000 and identical masks at 20, 25, 50, and 100 Hz. Patch lengths exceeding S raise an error rather than silently truncating.

5.1.3 Observability: say what could not be measured.

Rate-invariance would be dishonest without a second mechanism. A 20 Hz stream simply cannot report energy at 12 Hz, and a 0.4 s patch cannot resolve a 0.3 Hz band. Rather than emitting a small number that the encoder might read as “no energy here,” each token carries two masks. The *Nyquist observability* mask marks band k observable only if $c_k + 2\sigma_k \leq 0.9(r/2)$, a 10% guard below Nyquist so a band’s Gaussian tail does not straddle the aliasing edge; at 20 Hz this leaves 26 of 32 bands, at 25 Hz 27, and at 50 and 100 Hz all 32. The *resolution* mask marks band k resolved only if at least one full cycle fits in the patch, $c_k \geq 1/D$ for patch duration D ; a 0.5 s patch resolves 16 of 32 bands and a 1.5 s patch 25. A band that is present but blurry is flagged, not zeroed. Streams whose distributed form was resampled by their publisher take the Nyquist bound from the *acquisition* rate rather than the stored rate—upsampling cannot create information—which in our corpus applies to the 25 Hz earbud IMU stored on a 50 Hz grid.

5.1.4 Factored language conditioning. Tokens produced by a weight-shared tokenizer are agnostic to what produced them, so acquisition context must be injected. HALO factors that context in two, deliberately:

- a **role** string per channel that names *only* the axis—“x”, “y”, “z”—and is therefore constant across the entire corpus;
- a **sensor identity** string per sensor carrying device, modality, placement, and gravity convention, e.g. “a smartwatch accelerometer on the wrist; includes gravity”, or “a phone gyroscope on a trouser pocket”.

Both are embedded by the same frozen sentence encoder [58] and fused into the token, the sensor embedding through a gated residual initialised near zero so conditioning starts as a gentle bias and grows only if it earns its place. The factorisation is the point. Placement, device, and modality appear *only* in the sensor string and axis *only* in the role string, so no acquisition fact is injected twice when the two are summed. More importantly, the description a channel receives is now shared by every channel of that sensor and by every sensor of that kind across the corpus, which is precisely what makes

it too coarse to identify a dataset. Numeric metadata—sampling rate, patch duration—is deliberately *absent* from the text, because a frozen sentence encoder represents numbers unreliably; those values enter numerically through the bin→Hz map and the observability masks instead. During training the sensor strings are paraphrased and, with some probability, dropped to a generic string, so the encoder must tolerate an unfamiliar or missing description at deployment.

Whether the factorisation actually buys the anti-memorisation property it is designed for is an empirical question, not a design claim, and Section 6 tests it directly against a per-channel-text variant and a layout-locked variant (Tables 8 and 9).

5.1.5 Set encoder over physical time. Conditioned tokens enter a dual-branch transformer [73] that alternates *temporal* self-attention within a channel and *cross-channel* self-attention within a patch, with channel masks so absent or padded channels never contribute. Channels carry no positional index at all—their identity is text—so the encoder is permutation- and count-invariant over channels by construction, and a device with three channels and one with twelve are the same kind of input. Temporal positions use rotary embeddings [70] keyed to *physical time* in seconds rather than patch index, with periods spanning 0.4–600 s. This is the counterpart of the filterbank on the time axis: a 1.2 s gesture receives the same relative temporal code regardless of sampling rate or patch length, and because rotary positions are relative they remain exact under a key/value cache. The default configuration is $d=256$ with 6 layers and 8 heads (7.17M parameters), reading 6 s windows tokenised at two resolutions drawn per batch from 0.4–0.8 s and 1.0–1.5 s.

Streaming. Because the tokenizer uses no cross-patch statistics and the positions are relative, the only component mixing information across time is temporal self-attention, so offline and causal operation differ by nothing but the attention mask; the encoder implements both. We report this as a design affordance rather than a claim: this submission does not train a causal arm, distil offline to online, or measure streaming equivalence, and Section 7 lists it as future work.

5.2 Phase A: Label-Free Representation Pretraining

Phase A trains the tokenizer and encoder with objectives that never observe an activity label. This is a change of principle, not an economy: as long as the objective is a closed-vocabulary label prediction, the model is rewarded for exploiting whatever predicts labels within

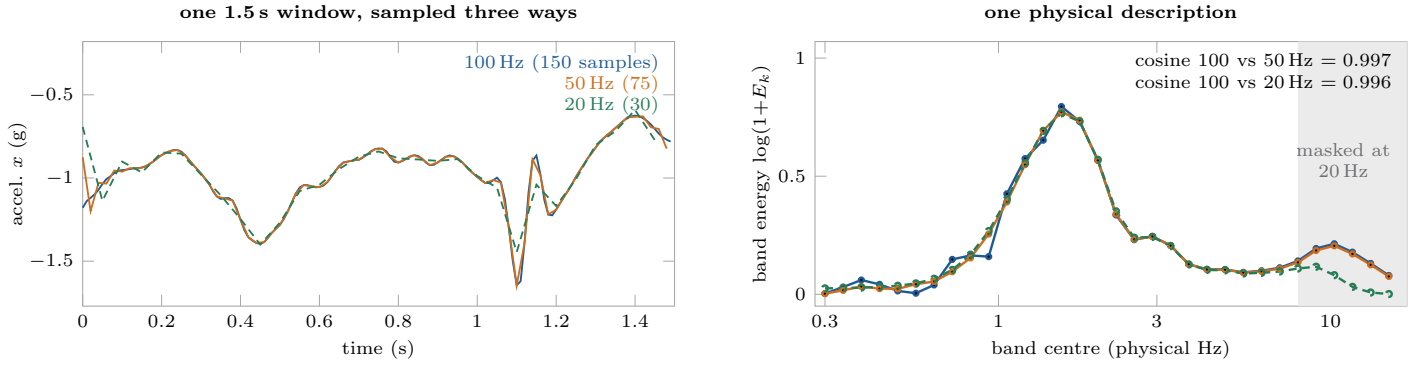


Figure 6: Rate-invariance, measured on a real window rather than argued. A single 1.5 s accelerometer patch of *walking* from the NFI-FARED wrist stream is anti-alias resampled to three rates and run through the tokenizer at each. *Left*: to a human the three are obviously the same motion, but to a model they are three different arrays—150, 75, and 30 numbers, no sample of one coinciding with a sample of another—so any encoder whose positional code is a timestep index sees three different structures. *Right*: the band energies land on top of one another, because a band denotes a physical frequency and the bin→Hz map has already absorbed the rate. Measured agreement is cosine 0.997 (100 vs 50 Hz) and 0.996 (100 vs 20 Hz) over the 26 bands all three rates can observe. The shaded region is what the 20 Hz stream physically cannot see: those six bands are *masked*, not estimated, so the encoder is told the content is missing instead of being handed a small number it would read as silence. Note that the 20 Hz curve does fall away inside the shaded band, which is the correct behaviour and precisely why the mask exists: an encoder that resampled the whole corpus to 20 Hz would silently take that loss on every stream, including the ones that measured the content perfectly well.

the training corpora, and Section 3.3 explained why that is the failure we most need to avoid. Removing labels from the loss also unlocks unlabelled and weakly labelled data, which is where the scale in wearables actually is.

The objective is a weighted sum of five terms plus one rail.

(A1) Masked prediction, with two targets. A validity-aware mask over physical time intervals hides a subset of tokens; the encoder predicts, at masked positions, both the *fixed* filterbank features of the hidden content and the *contextual latents* that a stop-gradient EMA teacher [2, 3, 25] produces from the unmasked window. The fixed target grounds the representation in physical quantities; the latent target lets the representation become more abstract than its own input features. The same cross-resolution mask is used for both so the two views cannot leak into each other, and the teacher is updated only after an optimizer step (decay 0.996) and is checkpointed with the model.

(A2) VICReg over augmented views. Two independently augmented views of the same window are pulled together under VICReg [5]—invariance, variance, and covariance terms reported separately—rather than under a contrastive loss. The reason is specific to this corpus: with NT-Xent, other windows in the batch are

declared negatives, and in a multi-dataset corpus the batch is full of windows of the same activity from other subjects and other datasets, so a contrastive objective actively pushes apart things that should be together. VICReg needs no negatives. SimCLR [14] and supervised contrastive [34] remain selectable as controls. Augmentations span jitter, scaling, time shift and warp, magnitude warp, window cropping and length jitter, rate change, channel dropout, joint $SO(3)$ rotation of co-located triads, gravity add/remove, and the text augmentations of Section 5.1.4. Text augmentation is applied with a *shared* seed across the two views, so that a view pair never differs only in whether its configuration was described—an asymmetry that would otherwise teach the encoder to ignore conditioning entirely.

(A3) Time–frequency consistency. An auxiliary time-domain encoder reads the raw samples of the same window, and its pooled output is pulled toward the main (frequency-domain) encoder’s under a separate VICReg head. This is inspired by TF-C [92] but is not faithful to it—there is no time-to-time contrastive term, and the time encoder is discarded at inference—so we describe it as a cross-domain consistency rail rather than as TF-C.

(A4) Cross-placement consistency. Where the corpus contains *verified simultaneous* recordings—the same

physical event captured at two or more body placements at the same instant—those pairs form an additional positive source at low weight and under a quota, so that the six-placement recordings cannot dominate. Positive identity comes from explicit physical-event identifiers written during conversion and carried into the pretraining index; it is never inferred from equal array lengths or row alignment, and never from an activity label. The training split contains 17,387 paired events over 55,270 paired windows.

(A5) A small physical grounding rail. Two self-derived, augmentation-aware targets (dominant cadence and a covariance eigenstructure ratio) are regressed at low weight. They use no labels and act as a sanity anchor rather than a driver.

Sampling is a per-window temperature draw with $P(\text{dataset}) \propto n^\alpha$, $\alpha=0.5$, which is the label-free counterpart of class balancing: a 144k-window free-living corpus must not swamp a 1k-window laboratory one, but neither should every source be forced to equal mass. Model selection uses subject-disjoint validation k -NN balanced accuracy—labels are used here, as a *metric*, never in the loss. Per-objective gradient norms, VICReg component values, minimum feature standard deviation, effective rank, and positive similarity by pair type are logged every step, and non-finite values or representation collapse fail the run rather than being smoothed over.

5.3 Phase B: The Evidence Engine

The encoder is frozen. What remains is to turn its representation into a decision over label strings the deployment supplies at run time—without the single sensor-to-text projection that Section 3.4 showed to be mis-specified.

Memory. Every labelled window available—from the training corpus and, optionally, from the target deployment—is encoded once and written to a versioned table. An entry stores the patch vector together with its label, subject, dataset and acquisition configuration, source window, physical event, sensor group, patch center time and duration, resolution, and validity. Sensor streams are encoded separately and a session’s patches form a query set. The table is append-only: adding a class, a placement, or a device is a write, not a training run. Every Phase-B artifact is bound to the exact Phase-A checkpoint, corpus, bank fingerprint, patch schema, and candidate-text vocabulary it was built from, and a behavioural probe rejects a bank whose encoder has since changed—silently stale memory is the most dangerous failure mode this design has.

Retrieval. Each query patch retrieves independently, by similarity in *sensor* space, under learned projections rather than hard dimension slices; each projection defines its own subspace with its own retrieval and its own evidence contribution, and head collapse is resisted by output decorrelation, projection diversity, and contribution-usage regularisation. An EMA copy of the projectors indexes memory while the online projectors score what comes back. During training the query’s own window, its physical event, and its subject are excluded from retrieval, and contributions are capped per source window and per label so a single long recording cannot dominate a vote. Evidence is aggregated with physical-duration normalisation within a resolution before resolutions are combined, so the denser short-patch grid does not receive an automatic vote-count advantage.

Candidate-aware decoding. Retrieved evidence is not simply averaged. A small set transformer contextualises query and evidence tokens together under learned structural roles (QUERY, EVIDENCE, CANDIDATE), then refines each evidence item’s label text as a residual in frozen sentence space—this is what lets a neighbour labelled *walking* but retrieved in a stair-climbing context contribute as *walking upstairs*—and pools evidence as a residual on the retrieval prior. Candidate tokens carry frozen label text plus a candidate role and attend permutation-equivariantly over the fixed evidence states, so neither candidate order nor evidence order carries meaning. Positions are window-relative continuous-time Fourier features, never a global sequence index, because relative time *across* windows is meaningless. Every sub-layer is gated by a LayerScale initialised near zero and the refinement and pooling residuals are zero-initialised, so the decoder at initialisation is exactly the untrained retrieval-plus-text-ensemble mechanism; any change it makes is attributable to training.

Confidence and rejection. The decoder emits non-negative evidence per candidate with attribution to the patches, sensors, subspaces, and retrieved examples that produced it, which is what allows a prediction to be inspected rather than merely reported. There is no UNKNOWN candidate; instead a rejection confidence is estimated separately [21, 62] from absolute evidence, retrieval density, patch and sensor agreement, normalised vote entropy, and the leading-candidate margin, and calibrated on truth-absent validation episodes with risk/coverage reported. Truth-present episodes require support, so always-reject is not a solution.

Episodic training. The decoder is trained on episodes that vary, independently: the candidate budget (10–100% of the vocabulary); the number of true-label memory exemplars (0, 1, 2, 4, 8, or all); the support available for competing labels; whether memory is same-configuration, cross-configuration only, or missing the query’s configuration entirely; the kind of distractors present (random, language-near, motion-family-near, physically confusable); and whether the truth is present at all. Complete label *families* are held out of decoder training and reserved for model selection, so candidate-text reasoning is always tested on concepts the decoder never fit.

Because the decoder is identity-at-initialisation, every trained arm has an exact untrained control, and Section 6 reports the two side by side rather than reporting the trained arm alone.

5.4 Deployment

A deployment is two pieces of configuration and no code. The *sensor layout* is one description string per sensor; the *candidate set* is a list of activity names. Both are embedded once by the frozen sentence encoder and cached, so run-time cost is one encoder pass plus a retrieval, with no per-inference text cost. Changing the label set, or moving to a different device, recomputes a cache.

The label-efficiency dial is the memory. At $k=0$ the target deployment contributes nothing and evidence comes from corpus exemplars and the candidate strings. At $k>0$ the user demonstrates each activity a few times; each demonstration costs one forward pass, is appended to the bank, and is immediately available—no gradient step, no retraining cycle, no revalidation of weights that did not move, and no requirement that the data leave the device. Where labels are plentiful, the same frozen features support a conventional supervised head. We expose this behind a small `predict()` interface and package the encoder for on-device execution (Core ML), demonstrated in the iOS case study of Section 6.5.

6 EXPERIMENTS

[TO BE COMPLETED] *The corpus, protocol, baseline roster, and table structure in this section are final and describe what is implemented. Result cells are reserved space, shown in orange, and are filled once the training run completes.*

6.1 Methodology

We ask four questions. **(Q1)** Along the label-efficiency curve— k target exemplars per class, for k from zero up to

full supervision—where does HALO stand against strong frozen encoders, and is there a regime it owns? **(Q2)** Does the tokenizer’s inductive bias do what Section 3.3 claims—is the representation genuinely rate-invariant, is the conditioning too coarse to identify a corpus, and, decisively, does anchoring let one flat memory serve devices it was not built from? **(Q3)** Which Phase-A objectives carry the transfer, and does any of them beat its own untrained control? **(Q4)** Does the system run on a phone, and does it behave in a real deployment?

6.1.1 Data. We train on 12 public HAR datasets contributing 20 distinct sensor streams (Table 3, top), and evaluate on 7 datasets never seen by any model during training (bottom). After conversion to native-rate grids and the removal of 95 windows whose values exceed physically possible sensor rails, the corpus is 343,140 non-overlapping six-second windows—572 hours of signal—over 93 activity labels, split subject-disjointly into 300,231 training and 42,909 validation windows. The validation split is constructed by a greedy label-cover procedure so that all 93 labels are represented.

6.1.2 Baselines and the fairness contract. Cross-model comparison in this setting is easy to get wrong, because the models do not accept the same inputs. We therefore state a single contract and apply it uniformly (Table 4). Each model occupies a *level* on each of four tiers—base architecture, sampling rate, channels/placement, and label vocabulary—and we never modify a released checkpoint’s input contract, because doing so discards the model being tested. Two baselines (CrossHAR, LiMUBERT) release no checkpoint compatible with our corpus, so we pretrain them ourselves on *our* training data with their own published objectives, which makes their data identical to ours; three (harnet, UniMTS, NormWear) are used as released, which is faithful but means their pretraining corpora differ from ours and from each other. This distinction is material—harnet is pretrained on roughly 700k person-days against our ~ 570 hours of windowed signal—and we state it wherever a comparison could be read as a like-for-like data-matched result. Closed-vocabulary models are bridged to arbitrary label strings with ConSE [53], which projects a classifier’s softmax over its training labels into the text-embedding space, so that “can predict an unseen label” is available to every model and is not something HALO is credited for.

6.1.3 Protocol. Scoring. Following the good-practice arguments of the zero-shot evaluation literature [78], every held-out dataset is scored against its *own* label

Dataset	Hz	#Str	Ch	#Win	Placement(s)
<i>Training corpus (12 datasets, 20 streams)</i>					
CAPTURE-24 [13]	100	1	3	144,142	wrist, free-living
WISDM [77]	20	2	6	70,201	pocket + wrist
XRF V2 [37]	50	6	6	34,764	pockets, wrists, glasses, earbuds
NFI-FARED [46]	100	2	6	26,520	wrist + lower back
HARMES [12]	50	1	6	18,576	wrist
UniMiB SHAR [48]	50	1	3	11,771	trouser pocket
UCI-HAR [1]	50	1	6	10,299	waist
HHAR [69]	50	1	6	9,587	waist
KU-HAR [66]	100	1	6	9,233	waist
SP-SW-HAR [45]	100	2	6	3,757	pocket + wrist
PAMAP2 [59]	100	1	6	3,186	wrist
MHEALTH [4]	50	1	6	1,104	wrist
<i>Held-out evaluation (7 datasets, never trained on)</i>					
MotionSense [44]	50	1	6	—	trouser pocket
RealWorld [71]	50	1	6	—	waist
Shoaib [64]	50	1	6	—	5 body positions
InclusiveHAR [31]	50	1	6	—	waist, mixed abilities
USC-HAD [89]	100	1	6	—	hip
TNDA-HAR [85]	50	1	6	—	wrist
UT-Complex [65]	50	1	6	—	wrist

Table 3: Corpus. *#Str* is the number of distinct sensor streams (one per device placement); *Ch* is the number of core channels reaching the model (3 = accelerometer only). Window counts are six-second windows on the native-rate grid after physical-plausibility filtering. InclusiveHAR comprises 10 able-bodied and 10 physically disabled subjects and is additionally reported with an ability-stratified breakdown. Six datasets were deliberately excluded from both splits. DSADS [6], HARTH [42], and Opportunity [61] use torso and limb IMU harnesses rather than phone or watch placements; RecGym [8] is distributed per-axis min–max normalised, which destroys the physical scale and gravity component the tokenizer depends on; MobiAct [74] was never materialised into grids; and HAPT [60] is excluded because it re-records the same 30 subjects as UCI-HAR [1], so including both would place the same people on both sides of a supposedly subject-disjoint boundary.

strings, never a shared training-wide ontology, so a prediction counts only if it retrieves that dataset’s exact activity name. Macro-F1 is primary, with accuracy alongside, because held-out label distributions are imbalanced.

Splits. All few-shot fits and all memory writes use subjects disjoint from those evaluated. Where a cohort is small enough that a single split is unstable, we rotate over grouped folds and report a confidence interval.

Label leakage. The label-text ensemble used at inference is purged of synonyms drawn from the evaluation datasets, so that no evaluation vocabulary is smuggled in through paraphrase. Retrieval hyper-parameters are selected on held-out *training* configurations, never on evaluation cells.

Model	Weights	Rate	Channels / placement	Labels
LiMU-BERT [80]	we pretrain	fixed 20 Hz	6-ch pad+mask; blind	ConSE
CrossHAR [27]	we pretrain	fixed 20 Hz	6-ch pad+mask; blind	ConSE
harnet [86]	released	fixed 30 Hz	3-ch acc; wrist-implied	ConSE
UniMTS [91]	released	resampled	joint-graph placement, 3-ch	native text
NormWear [43]	released	fixed 65 Hz	variable ch + mask; blind	native text
HALO (ours)	we train	native	per-sensor language	native

Table 4: Baselines and where each sits on the heterogeneity stack. “Blind” means the model receives no information about where the sensor was worn. *Weights:* two baselines are pretrained by us on our corpus with their published objectives (data-matched); three are used as released (faithful, but pretrained on their own, larger, corpora—harnet on ~700k person-days). We report both groups separately rather than averaging over them.

Parity control. To separate the architecture from its native-rate and rich-text inputs, HALO is additionally evaluated under baseline-equivalent inputs: common 20 Hz resampling and a neutral sensor description (Table 10).

6.2 Q1: The Label-Efficiency Curve

This is the headline experiment. For each held-out dataset and each budget k of labelled target windows per class, we compare three mechanisms on identical data: HALO’s memory, given the k exemplars as an append with *no gradient update*; a probe fitted on the same k exemplars over HALO’s frozen features; and the same probe over frozen harnet features, which is the strong-representation control. We additionally report *retrieval over harnet features*—the same memory mechanism on a different encoder—because without it a win could be attributed to the mechanism when it belongs to the representation, or the reverse.

Pre-registered hypothesis and kill criterion. We expect retrieval to win at small k , where a nearest-neighbour decision beats a linear fit on a handful of points, and the probe to overtake somewhere between $k=10$ and $k=50$. If that holds, the regime HALO owns is $k \leq 10$, which is exactly the enrollment setting a user can perform in under a minute. If the probe dominates at *every* k including $k=1$, retrieval has no regime of its own; we will report that outcome plainly.

6.3 Q2: Does the Inductive Bias Do What It Claims?

Section 3.3 made two falsifiable claims about the tokenizer. Both are tested directly rather than inferred from end-task accuracy.

Mechanism	labelled target windows per class (k)					
	0	1	5	10	50	full
HALO memory (no gradient step)	■	■	■	■	■	■
HALO frozen + probe	■	■	■	■	■	■
harnet frozen + probe	■	■	■	■	■	■
harnet frozen + memory [†]	■	■	■	■	■	■
UniMTS (native text)	■	■	■	■	■	■
CrossHAR + ConSE	■	■	■	■	■	■
LiMU-BERT + ConSE	■	■	■	■	■	■
NormWear (native text)	■	■	■	■	■	■

Table 5: Macro-F1 versus label budget, averaged over the 7 held-out datasets, subject-disjoint throughout. $k=0$ is the zero-shot point and $k=full$ the supervised one; they are ends of one curve, not separate experiments, and every row uses *one* frozen encoder across all k . [†] The mechanism-versus-representation control: if retrieval helps only on HALO features, the contribution is the encoder; if it helps equally on harnet’s, it is the memory. We pre-register that we will report this control regardless of which way it falls.

Dataset	$k=0$		$k=10$	
	Acc	F1	Acc	F1
MotionSense	■	■	■	■
RealWorld	■	■	■	■
Shoaib	■	■	■	■
InclusiveHAR	■	■	■	■
<i>able-bodied</i>	■	■	■	■
<i>disabled</i>	■	■	■	■
USC-HAD	■	■	■	■
TNDA-HAR	■	■	■	■
UT-Complex	■	■	■	■
Mean	■	■	■	■

Table 6: Per-dataset HALO performance at the two ends of the enrollment regime. InclusiveHAR is reported stratified by ability, because a model that works only for able-bodied users is not a foundation for rehabilitation applications.

Rate-invariance is measured, not assumed. A closed-form check confirms that a pure tone deposits identical band energies at 20, 50, and 100 Hz, and that changing the DFT padding from $S=256$ to 512 leaves band energies and masks unchanged. The end-task version of the claim is Table 8: the same encoder trained on the same data but reading a common 20 Hz resampling of every stream. This is the experiment that distinguishes a physical-frequency *anchoring* from mere resolution flexibility—if the 50 and 100 Hz streams carry discriminative content above 10 Hz, resampling must destroy it, and the gap is the measurement of how much.

Does anchoring actually make one cross-device memory well-posed? This is the experiment that decides whether HALO is an assembly of known parts or an enabling result, so we state it precisely. A flat memory shared across devices is only meaningful if a query encoded from one configuration retrieves useful neighbours stored from a *different* configuration. Table 7 measures exactly that: rows are the configuration a query comes from, columns the configuration the memory is restricted to, and cells report retrieval quality. The diagonal is same-configuration retrieval, which every system can do; the off-diagonal is the claim.

The measurement is unusually clean because parts of our corpus contain *verified simultaneous* recordings—the same person performing the same activity at the same instant, captured by two or more devices at different placements (XRF V2 at six placements, SP-SW-HAR phone and watch, NFI-FARED wrist and lower back, WISDM pocket and watch). Querying with one stream against a memory built only from its simultaneous partner isolates acquisition configuration from subject, session, and activity entirely: any degradation off the diagonal is attributable to the configuration change and to nothing else. We report the same matrix for a resample-to-common-rate variant of our own encoder, and for the frozen baselines, so that the comparison is of mechanisms rather than of corpora.

Is the naive assembly enough? The components HALO uses have individually appeared before: variable channel counts, variable rates, and language-mediated label interfaces all have prior solutions. The claim that they must be assembled *this particular way* is therefore an empirical one, and Table 8 includes the obvious alternative assembly as a row: resample every stream to a common rate, attach free text per channel, and put the same retrieval mechanism on top. If that arm matches HALO, the architecture is a combination and we will say so; the mutual-entailment argument of Section 2.1 predicts it will not, and predicts a specific reason—the per-channel conditioning should show up as elevated dataset identifiability in Table 9, and the resampling should show up as a rate gap in Table 8.

Conditioning must not be a corpus fingerprint. Table 9 reports a dataset-identity probe: a linear classifier fitted on frozen embeddings to predict which *source dataset* a window came from, on held-out subjects. A representation that has absorbed the acquisition description as a corpus identifier will make this easy; one whose conditioning is genuinely compositional will not. We report it for HALO’s factored per-sensor conditioning, for a per-channel free-text variant, and for the baselines, and

Query → memory	HALO		resample-to-20 Hz	
	same	cross	same	cross
wrist → pocket (SP-SW-HAR)	■	■	■	■
wrist → lower back (NFI)	■	■	■	■
pocket → watch (WISDM)	■	■	■	■
wrist → ear/glasses (XRF V2)	■	■	■	■
100 Hz → 20 Hz memory	■	■	■	■
cross/same retention		■		■

Table 7: Cross-configuration retrieval: the enabling claim. Each row queries with one acquisition configuration against a memory containing *only* a different one. Because these pairs are verified simultaneous recordings—same subject, same activity, same instant—the only thing that varies is the configuration, so off-diagonal degradation is attributable to it alone. *Retention* is cross divided by same: a system whose features denote the same physical quantity across devices should retain most of its quality, while one that re-samples or conditions on the rate should not. This is what a flat, device-agnostic memory requires, and it is the property that no prior retrieval-based HAR system has needed to have.

Tokenizer configuration	k=0	k=10	rate gap [‡]
Physical-Hz filterbank, sensor-grouped, factored text	■	■	■
– filterbank (resample all streams to 20 Hz)	■	■	■
– sensor grouping (per-channel free text)	■	■	■
– language conditioning (layout-locked input)	■	■	■
– observability / resolution masks	■	■	■
– signed DC (gravity) feature	■	■	■
<i>naïve assembly</i> : resample + per-channel text	■	■	■

Table 8: Tokenizer inductive-bias ablation (macro-F1 over the 7 held-out datasets), each component removed independently. The last row is not an ablation but a *rival assembly*: the same retrieval mechanism over an encoder built the conventional way—every stream resampled to a common rate, free text attached per channel. It is the arm that tests whether HALO is a combination of known parts. [‡] *Rate gap* is performance on the 100 Hz held-out cell minus performance on the same data downsampled to 20 Hz—the direct measurement of what resampling costs.

we report it whether or not it favours us. It is cheap—frozen features and a linear fit—and it is the only direct evidence we have for the “too little bias” failure mode.

6.4 Q3: Which Phase-A Objectives Carry the Transfer?

Each Phase-A term is removed independently with all others held fixed (Table 11). Two controls matter more

Encoder	dataset-ID acc. ↓	activity F1 ↑
HALO (factored per-sensor text)	■	■
HALO (per-channel free text)	■	■
HALO (no language conditioning)	■	■
harnet [86]	■	■
UniMTS [91]	■	■
chance	■	—

Table 9: Dataset-identity probe. A linear classifier predicting the *source dataset* from frozen embeddings of held-out subjects. High accuracy here means the representation encodes corpus provenance—the “too little inductive bias” failure of Section 3.3—and predicts poor cross-dataset transfer. Lower is better; the activity column confirms the representation has not simply been degraded.

Configuration	k=0	k=10
HALO (native Hz, per-sensor text)	■	■
HALO <i>parity</i> (20 Hz, neutral text)	■	■
Best baseline (data-matched group)	■	■
Best baseline (released-checkpoint group)	■	■

Table 10: Parity control: HALO under its native inputs versus baseline-equivalent inputs, so that any remaining margin is attributable to the architecture rather than to input advantages. Baselines are split by whether their weights were trained on our corpus or released by their authors.

than the ablation itself. First, the *untrained control*: the evidence decoder is identity-at-initialisation by construction, so for every trained arm we report the same architecture untrained; a trained component that does not beat its own initialisation is reported as such. Second, the *label-supervised control*: the original supervised recipe (masked prediction + label-SupCon + grounding rail) remains selectable, so the claim that going fully label-free helps is measured, not asserted.

6.5 Q4: On-Device Cost and a Real Deployment

[TO BE COMPLETED] *On-device latency, peak memory, retrieval cost as a function of memory-bank size, and the case-study numbers, once the encoder is compiled and profiled.*

We implement the full pipeline as an iOS application and will report: encoder latency and peak memory under Core ML; retrieval latency as a function of bank size; and a continuous case study in which a user enrolls

Phase-A configuration	$k=0$	$k=10$
Full (A1 fixed + EMA, VICReg, TF, placement, rail)	—	—
– EMA-latent target (fixed A1 only)	—	—
– augmentation VICReg	—	—
– time-frequency consistency	—	—
– cross-placement positives	—	—
– physical grounding rail	—	—
VICReg → NT-Xent (contrastive control)	—	—
Label-supervised recipe (SupCon; <i>uses labels</i>)	—	—

Table 11: Phase-A objective ablation, macro-F1 over the 7 held-out datasets. The last two rows are controls, not variants: they test whether the non-contrastive and label-free choices were worth making.

Reserved for the case-study figure: enrollment by demonstration, the mid-session placement change, and the streamed prediction trace.

Figure 7: [TO BE COMPLETED] Deployment setting of the smartphone case study.

activities by demonstration, changes the device’s body placement mid-session, and introduces activity names that were never enrolled. That case study is the deployment form of the label-efficiency argument: enrollment by demonstration is what $k \leq 10$ means to a user.

7 CONCLUSION AND DISCUSSION

We argued that the obstacle to a reusable IMU foundation model is not a shortage of data but two structural facts about the modality. Inertial signals admit no closed grammar: the map from a motion to a recording is composed at deployment time from continuous, composable nuisance factors, so cross-configuration compatibility has to be supplied as calibrated inductive bias rather than accumulated as coverage. And sensor similarity does not track linguistic similarity—measured on our own vocabulary, the ordering is not merely noisy but inverted—so projecting sensor features into a sentence-embedding space is a mis-specified target that rewards dataset memorisation. HALO responds with a tokenizer that commits to the physics that does not vary (rigid sensor triads, a physical-frequency axis, signed gravity) and defers everything that does to per-sensor language and per-token observability masks; and with an evidence engine that retrieves in sensor space and votes in label space, so the

two geometries never meet in one projection. Representation pretraining uses no activity labels at all; labels enter only as annotations on retrieved memory entries, which is what makes the amount of target supervision a dial rather than a mode.

What we are not claiming. Three things, stated plainly so they are not read into the work. We do not claim that accepting a variable number or order of channels is a contribution; padding with a mask is one line of preprocessing, and our own corpus spans only three to six channels per device. We do not claim that recognising unseen label *strings* is a contribution; ConSE has done that since 2014, which is exactly why we equip every closed-vocabulary baseline with it. And where labels are plentiful and a single fixed configuration is known in advance, we do not expect retrieval to beat supervised fitting—that is a solved problem and the honest place to concede it is here rather than in a rebuttal.

Limitations. Our training corpus is roughly 570 hours of windowed signal, several orders of magnitude smaller than the largest released accelerometer models, and no design choice compensates for that on tasks where scale is what matters. The evidence engine’s ceiling is its encoder: if retrieval returns neighbours of the wrong activity, no decoder recovers, and we measure retrieval purity precisely because it bounds everything downstream. The cross-placement objective depends on datasets with verified simultaneous recordings, of which there are few. Windows are single-activity, so the model learns to recognise segments and not to find boundaries. And the encoder supports causal operation but this submission does not train or measure it, so streaming remains a design affordance rather than a demonstrated capability.

Future work. Four directions follow directly. *Dense supervision*: per-frame labels from concatenated or free-living streams would teach onset and transition detection, which single-activity windows cannot. *Streaming, properly*: train a causal arm, distil offline to online, and verify streaming equivalence, converting the affordance into a claim. *Scale on the axis that is actually free*: because Phase A needs no labels, the corpus can grow into unlabelled free-living data, which is where the hours are. *Controlled-shift evaluation*: the corpus contains simultaneous multi-placement recordings of the same person at the same instant, which isolate placement from subject, session, and activity—a cleanly controlled axis that a blended cross-dataset average hides, and the right way to measure the compatibility claim one nuisance factor at a time.

Finally, a note on what a negative result would mean. The label-efficiency curve (Section 6.2) has a pre-registered kill criterion: if a linear probe on frozen

features dominates retrieval at every budget including a single example per class, then the non-parametric mechanism has no regime of its own and the argument reduces to its operational properties—one encoder across many deployments, append-only updates without revalidation, on-device enrollment, and inspectable decisions. Those are real, but they are engineering properties, and we would say so rather than reframe the metric.

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